

SUDHANVA DEVANATHAN

Boston, MA

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Experience

BlackRock

Mumbai, India

Software Engineer II (Associate)

Jan 2021 – July 2024

- Architected a scalable **Python** orchestration pipeline handling **50+ million rows** for SEC regulatory analytics, optimizing **Sybase/SQL** performance to achieve a **30% improvement** in processing speed.
- Designed a fault-tolerant **Java/Spring Boot** microservice for centralized configuration, utilizing **gRPC** and **Protobuf** to decouple DB dependencies and reduce connection overhead by **80%**.
- Architected an event-driven monitoring ecosystem using **Python asyncio**, deploying distributed servers and pollers to reduce detection latency by **40%** for critical financial models.
- Modernized the CI/CD ecosystem by migrating pipelines from **Jenkins** to **Azure DevOps**, integrating automated **JUnit** suites to increase code coverage to **95%** and reduce deployment failures.
- Fostered technical growth teammates through thoughtful code reviews, sharing **clean design methodologies** that improved code quality and minimized technical debt across the team.

BlackRock

Mumbai, India

Software Engineer I (Analyst)

July 2018 – Dec 2020

- Engineered a memory-aware scheduling algorithm to enable previously unfeasible high-load analytics; implemented dynamic routing to optimal nodes, resolving resource bottlenecks and reducing latency by **20%**.
- Contributed to the architectural rewrite of a legacy Risk Calculator into **Go (Golang)**, implementing complex data orchestration workflows using a custom **Domain Specific Language (DSL)**.
- Collaborated with Product Managers to build an automated quality control system for environment readiness, reducing manual data validation time by **30%** and improving accuracy by **25%**.

Projects

Distributed Key-Value Store with Raft Consensus | *Go, gRPC, Protobuf*

Dec 2025

- Built a strongly consistent, distributed key-value store in **Go**, implementing the **Raft consensus algorithm** for leader election and log replication across a 5-node cluster.
- Implemented **sharding** using **consistent hashing** to distribute data evenly and ensure high availability during node failures, mirroring architecture patterns found in Cassandra.
- Engineered a persistent storage layer using a **Write-Ahead Log (WAL)** and an on-disk **LSM Tree**, ensuring data durability and crash recovery in under 2 seconds.
- Optimized system performance by implementing **Log Compaction** (Snapshotting), preventing unbounded log growth and reducing memory footprint by **60%** during long-running sessions.

High-Availability Layer 7 Load Balancer | *Go, Redis, Docker*

Dec 2025

- Engineered a high-performance Layer 7 Load Balancer in **Go**, integrating **Redis** for shared state management and **Docker** for containerization to support dynamic algorithm switching.
- Implemented advanced reliability patterns including **idempotent request processing** and **automatic retries with exponential backoff**, ensuring data consistency during transient network failures.

Deep Reinforcement Learning for Atari Games | *Python, PyTorch*

Apr 2025

- Implemented Deep Reinforcement Learning algorithms (DQN, PPO) to train agents on Pong, achieving super-human performance.
- Built modular training pipelines with **GPU acceleration** and checkpointing to evaluate policy robustness.

Education

Northeastern University

Boston, MA

Master of Science in Computer Science; GPA: 3.87

Sep 2024 – Apr 2026

- **Relevant Coursework:** Distributed Systems, Reinforcement Learning, Algorithms, Design Paradigms, Advanced Algorithms

Technical Skills

Languages: Java, Python, Go (Golang), SQL, C++, Perl, Shell Scripting

Core Concepts: Distributed Systems, Microservices, Event-Driven Architecture, REST APIs, CAP Theorem

Frameworks: Spring Boot, Hibernate, gRPC, Protobuf, Kafka, PyTorch, Pandas, NumPy

Cloud & DevOps: AWS (EC2, Lambda, S3), Docker, Kubernetes, Azure DevOps, Jenkins, CI/CD

Databases: Redis, Cassandra (NoSQL), Sybase, PostgreSQL, DynamoDB